

MODEL QUESTION PAPER

REVISION-2010

SUB CODE:3053

Reg. No.....

Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN ELECTRICAL & ELECTRONICS

ENGINEERING, OCTOBER 2011

ELECTRONIC DEVICES AND CIRCUITS

Time :3hours

Maximum marks:100

PART A

(maximum marks **10**)

I. Answer the following questions in 2 or 3 sentences:

marks

1. Define the term barrier potential ?
2. Calculate I_E in a transistor for which $\beta = 50$ and $I_B = 20\mu A$?
3. Define the bandwidth in an amplifier.
4. List out two methods of amplifier coupling .
5. What is the Barkhausen criteria for oscillation?

(5x2=10)

PART B

(Maximum marks :30)

II Answer any five questions from the following

1. Analyse the DC load line concept of a transistor.
2. Explain the working of half wave rectifier .
3. Explain the capacitor filter circuits with wave forms.
4. Explain the negative feed back with a circuit diagram.
5. Draw the monostable multivibrator circuit using IC555.
6. Name the various types of tuned oscillators.
7. Differentiate an ordinary diode and zener diode

(5x6 =30)

PART C

(Answer any four full questions, selecting one from each unit)

UNIT I

- III. a) Analyse the behavior of PN junction in forward biasing 8
b) Describe the current flow in a NPN transistor. 7

OR

- IV a) Explain any one method of PN junction formation. 7
b) Analyse the common base configuration of a transistor. 8

UNIT II

- V Explain the construction and working of a centre tap full wave rectifier with necessary wave forms. 15

OR

- VI a) Describe the regulator circuit using 7905 IC. 7
b) Explain the working of a biased clipping circuit with wave forms. 8

UNIT III

- VII a) Explain the working of a class C amplifier 8
b). Describe the various schemes of coupling of amplifiers 7

OR

- VIII a) Describe the positive feed back and its applications 7
b) Explain the working of transformer coupled amplifier with a neat diagram 8

UNIT IV

- IX a) Explain the working principle of an oscillator. 8
b) Describe the working of Colpitts oscillator. 7

OR

- X a) Explain the operation of a crystal oscillator. 8
b) Explain the terms UTP and LTP of a Schmitt trigger. 7
